

SHISHMAREF, AK, USA

WMO: 701195

Lat: 66.250N

Lon: 166.089W

Elev: 2

StdP: 101.30

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26625

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-34.0	-31.9	-34.0	0.2	-26.9	-33.5	0.2	-27.4	15.9	-17.3	14.4	-14.2	3.8	250	0.974

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.4	16.0	13.1	14.1	11.7	12.9	10.9	13.5	15.6	12.2	14.0	11.2	13.0	4.7	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	8.9	15.0	11.2	8.3	13.8	9.9	7.6	12.7	37.6	15.5	34.6	14.4	32.0	13.0	17.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 14.3	12.4	11.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-32.0	14.7	1.1	1.3	-32.9	15.6	-33.5	16.4	-34.2	17.1	-35.0	18.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.9	-17.2	-17.6	-18.2	-11.3	-1.4	4.3	9.8	9.4	6.0	-0.9	-8.0	-14.8
	DBStd	11.72	7.99	7.92	6.75	6.43	4.11	3.48	2.70	2.42	2.74	3.78	5.69	7.02
	HDD10.0	5506	843	774	874	638	354	174	36	41	122	339	542	769
	HDD18.3	8490	1102	1007	1132	888	612	419	265	278	369	597	792	1028
	CDD10.0	57	0	0	0	0	0	4	29	21	3	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	5.2	5.3	5.5	4.9	4.8	4.0	4.0	4.7	5.4	5.7	5.7	6.0	5.9
(15) Precipitation	PrecAvg	578	41	45	31	30	29	36	67	83	58	64	52	41
	PrecMax	785	92	141	86	80	74	70	122	176	123	111	143	74
	PrecMin	413	2	11	4	8	4	9	22	23	13	15	7	1
	PrecStd	102	21	30	23	17	19	13	25	37	26	28	34	19
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.3	1.2	1.1	2.5	7.7	16.1	18.0	17.5	13.7	7.4	2.4	2.2
		MCWB	0.3	-0.1	0.0	1.1	5.1	13.0	14.1	14.1	10.7	5.2	1.4	0.1
	2%	DB	-1.2	-0.9	-2.0	2.0	5.0	12.9	16.3	15.2	12.3	6.1	1.3	0.0
		MCWB	-1.7	-1.3	-2.7	0.9	3.2	10.6	13.2	12.6	10.3	4.3	0.5	-0.7
	5%	DB	-2.8	-2.8	-3.9	0.9	3.7	11.3	14.8	13.8	11.2	4.8	0.9	-2.4
		MCWB	-3.7	-3.7	-4.9	-0.2	2.3	9.2	12.3	11.7	9.5	3.4	0.0	-3.1
	10%	DB	-5.8	-5.9	-7.6	-1.3	2.7	9.1	13.7	12.7	9.9	3.8	-1.0	-4.6
		MCWB	-6.4	-6.5	-8.3	-1.6	1.5	7.5	11.3	10.9	8.4	2.5	-1.5	-5.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.1	0.0	1.1	5.3	13.3	14.9	14.7	11.3	5.7	1.6	0.5
		MCDB	1.1	0.9	0.8	2.2	7.3	15.5	17.2	17.0	12.6	6.9	1.9	1.4
	2%	WB	-1.8	-1.2	-2.4	0.4	3.5	10.8	13.8	13.3	10.4	4.6	0.6	-0.9
		MCDB	-1.5	-0.8	-2.0	1.4	4.8	12.8	16.0	15.3	11.9	5.9	1.2	-0.2
	5%	WB	-3.7	-3.6	-4.5	-0.3	2.5	9.2	12.7	12.0	9.5	3.6	-0.2	-3.1
		MCDB	-2.7	-2.6	-3.5	0.6	3.6	11.1	14.7	13.5	11.0	4.7	0.5	-2.4
	10%	WB	-5.9	-5.9	-8.1	-2.0	1.7	7.7	11.5	11.1	8.2	2.7	-1.4	-5.1
		MCDB	-5.3	-5.3	-7.5	-1.7	2.7	9.1	13.4	12.8	9.3	3.7	-1.0	-4.3
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.0	7.8	7.5	4.8	4.6	4.4	3.8	3.4	3.3	4.9	6.0
		MCDBR	6.8	9.1	9.0	5.6	5.7	7.1	6.5	5.5	4.7	3.7	3.6	7.5
		MCWBR	6.7	8.9	8.6	5.1	4.7	5.1	4.3	3.9	3.3	3.1	3.5	7.3
	5% WB	MCDBR	7.0	9.1	9.0	6.2	5.5	6.8	5.8	5.1	4.4	3.6	3.7	7.0
		MCWBR	6.9	9.0	8.6	5.8	4.5	5.0	4.2	3.8	3.2	3.1	3.7	6.8
(40) Clear-Sky Solar Irradiance	taub		0.168	0.245	0.268	0.336	0.369	0.351	0.365	0.384	0.315	0.276	0.364	0.178
	taud		1.904	2.100	2.182	2.006	2.069	2.400	2.441	2.330	2.513	2.297	2.805	1.905
	Ebn at Noon		272	614	798	806	819	858	832	764	759	580	262	28
	Edh at Noon		16	54	83	130	136	101	95	96	63	45	19	3
(44) All-Sky Solar Radiation	RadAvg		0.09	0.68	2.45	4.54	5.51	5.81	4.57	3.36	2.07	0.73	0.14	0.01
	RadStd		0.01	0.08	0.19	0.27	0.61	0.48	0.40	0.36	0.20	0.07	0.02	0.00

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+1.12	N/A	+1.10	+0.88	+0.64	N/A	-342	-412	+45	N/A	

Nomenclature: See separate page